CHLOROFORM Trichloromethane Methane trichloride Formyl trichloride	ICSC: 0027 (April 2024)
CAS #: 67-66-3 **UN #: 1888** **EC Number: 200-663-8**	

	ACUTE HAZARDS	PREVENTION	FIRE FIGHTING
FIRE & EXPLOSION	Not combustible. See Notes. Gives off irritating or toxic fumes (or gases) in a fire. Risk of fire and explosion on contact with incompatible substances. See Chemical Dangers.	NO contact with incompatible substances. See Chemical Dangers.	In case of fire in the surroundings, use appropriate extinguishing media. In case of fire: keep drums, etc., cool by spraying with water.

STRICT HYGIENE! AVOID EXPOSURE OF ADOLESCENTS AND CHILDREN!

	SYMPTOMS	PREVENTION	FIRST AID
Inhalation	Cough. Shortness of breath. Dizziness. Drowsiness. Headache. Nausea. Unconsciousness.	Use local exhaust or breathing protection.	Fresh air, rest. Artificial respiration may be needed. Refer for medical attention.
Skin	MAY BE ABSORBED! Redness. Pain. Dry skin.	Protective gloves. Protective clothing.	Remove contaminated clothes. See Notes. Rinse skin with plenty of water or shower. Refer for medical attention .
Eyes	Redness. Pain.	Wear face shield or eye protection in combination with breathing protection.	First rinse with plenty of water for several minutes (remove contact lenses if easily possible), then refer for medical attention.
Ingestion	Abdominal pain. Nausea. Vomiting. Further see Inhalation.	Do not eat, drink, or smoke during work.	Do NOT induce vomiting. Rinse mouth. Give nothing to drink. Refer for medical attention .

SPILLAGE DISPOSAL

Evacuate danger area! Consult an expert! Personal protection: complete protective clothing including self-contained breathing apparatus. Do NOT let this chemical enter the environment. Collect leaking liquid in sealable containers. Absorb remaining liquid in sand or inert absorbent. Then store and dispose of according to local regulations.

STORAGE

Store only in original container. Separated from food and feedstuffs and incompatible materials. See Chemical Dangers. Ventilation along the floor. Store in an area without drain or sewer access.

PACKAGING

Put breakable packaging into closed unbreakable container. Do not transport with food and feedstuffs.

CLASSIFICATION & LABELLING

According to UN GHS Criteria

DANGER

Harmful if swallowed or if inhaled
Causes skin and eye irritation
May cause drowsiness or dizziness
Suspected of causing cancer
Suspected of damaging fertility or the unborn child
Causes damage to the liver, the kidneys and the respiratory tract through prolonged or repeated exposure
Harmful to aquatic life

Transportation
UN Classification
UN Hazard Class: 6.1; UN Pack Group: III

International Labour Organization World Health Organization

Prepared by an international group of experts on behalf of ILO and WHO, with the financial assistance of the European Commission.

European Commission

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PHYSICAL & CHEMICAL INFORMATION

Physical State; Appearance
VOLATILE COLOURLESS LIQUID WITH CHARACTERISTIC ODOUR.

Physical dangers
The vapour is heavier than air and may accumulate in lowered spaces causing a deficiency of oxygen.

Chemical dangers
Decomposes on heating. This produces toxic and corrosive fumes of hydrogen chloride (see ICSC 0163), phosgene (see ICSC 0007) and chlorine (see ICSC 0126). Reacts violently with strong bases, strong oxidants, alkalis, acetone and some metals such as aluminium, magnesium and zinc. This generates fire and explosion hazard. Attacks plastics, rubber and coatings.

Formula: $CHCl_3$
Molecular mass: 119.4
Boiling point: 62°C
Melting point: -64°C
Density (at 20°C): 1.48 g/ml
Solubility in water, g/100ml at 20°C: 0.8 (slightly soluble)
Vapour pressure, kPa at 20°C: 21.2
Relative vapour density (air = 1): 4.12
Relative density of the vapour/air-mixture at 20°C (air = 1): 1.7
Octanol/water partition coefficient as log Pow: 1.97

EXPOSURE & HEALTH EFFECTS

Routes of exposure
The substance can be absorbed into the body by inhalation, through the skin and by ingestion.

Effects of short-term exposure
The substance is irritating to the eyes and skin. The substance may cause effects on the central nervous system. This may result in central nervous system disorders and depression. The substance may cause effects on the liver and kidneys at high concentrations. This may result in liver impairment and kidney damage. The effects may be delayed. See Notes. Medical observation is indicated.

Inhalation risk
A harmful contamination of the air can be reached very quickly on evaporation of this substance at 20°C.

Effects of long-term or repeated exposure
The substance defats the skin, which may cause dryness or cracking. The substance may have effects on the nasal passage. This may result in tissue lesions. The substance may have effects on the central nervous system, liver and kidneys. This may result in impaired functions. This substance is possibly carcinogenic to humans. Animal tests show that this substance possibly causes toxicity to human reproduction or development.

OCCUPATIONAL EXPOSURE LIMITS

TLV: 10 ppm as TWA; A3 (confirmed animal carcinogen with unknown relevance to humans).
EU-OEL: 10 mg/m^3, 2 ppm as TWA; (skin)

ENVIRONMENT

The substance is harmful to aquatic organisms.

NOTES

Turns combustible on addition of small amounts of a flammable substance or an increase in the oxygen content of the air.
Isolate contaminated clothing by sealing in a bag or other container.
Use of alcoholic beverages enhances the harmful effect.
Depending on the degree of exposure, periodic medical examination is suggested.
The odour warning when the exposure limit value is exceeded is insufficient.
The toxicity of this substance is driven by the metabolite phosgene (See ICSC 0007).

ADDITIONAL INFORMATION

EC Classification